

Introduction to Astronomical Photometry

M1 Astro-CCP HAP 703P

Atelier Astrophysique Observationnelle 1

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LUPM LABORATOIRE MONTPELLIER UNIVERS & PARTICULES

Outline

1 Spectral energy distributions of astrophysical sources

2 Astronomical photometry

3 Astronomical photometry: applications

4 Bibliographic ressources

Outline

1 Spectral energy distributions of astrophysical sources

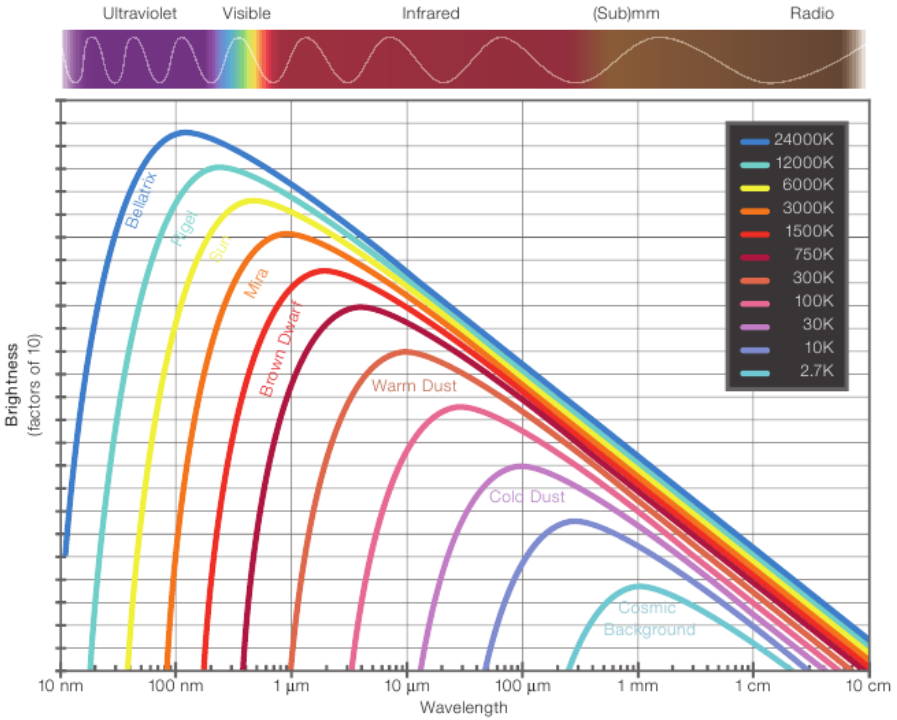
2 Astronomical photometry

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SEDs of astrophysical sources

- black-body radiation
  - ideal thermal radiation
    - emittance $(\mathrm{W.m^{-2}})$
    - wavelength of peak emission
  - ➔ temperature
- stellar SEDs
  - approximately black bodies
  - absorption + emission lines
  - atoms + molecules
- SEDs of galaxies and quasars
  - composite of stars, nebulae, ...
  - effect of redshift

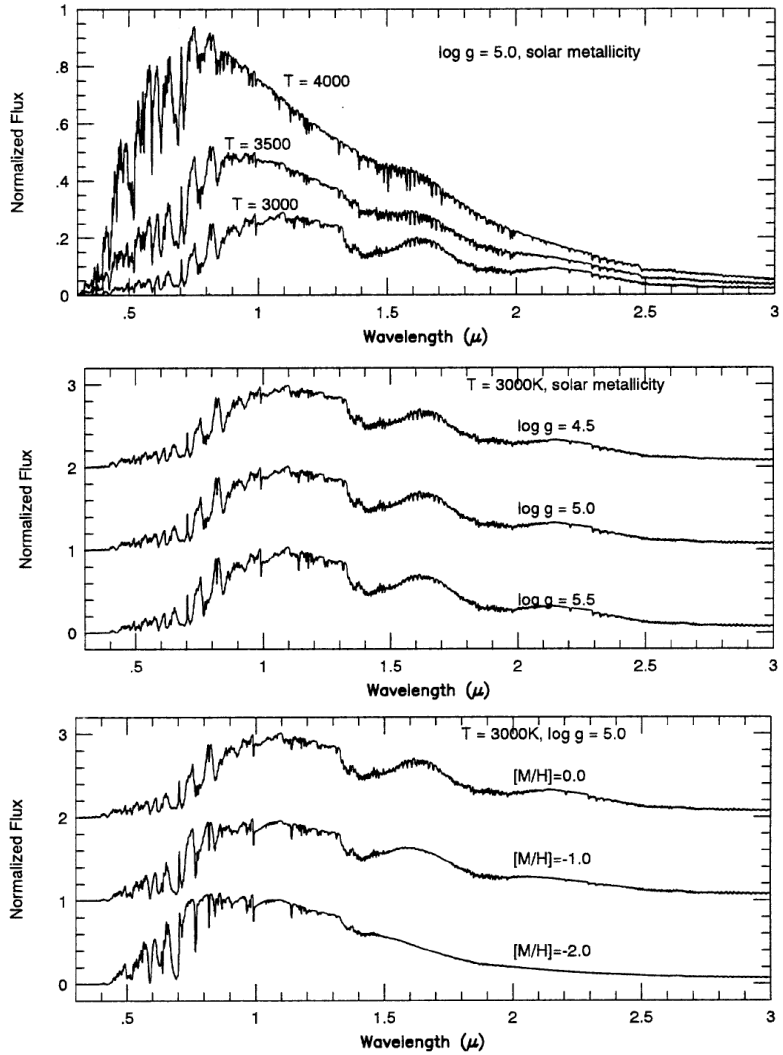


Blackbody SED as a function of temperature

Credit: Robert Hurt/ESO

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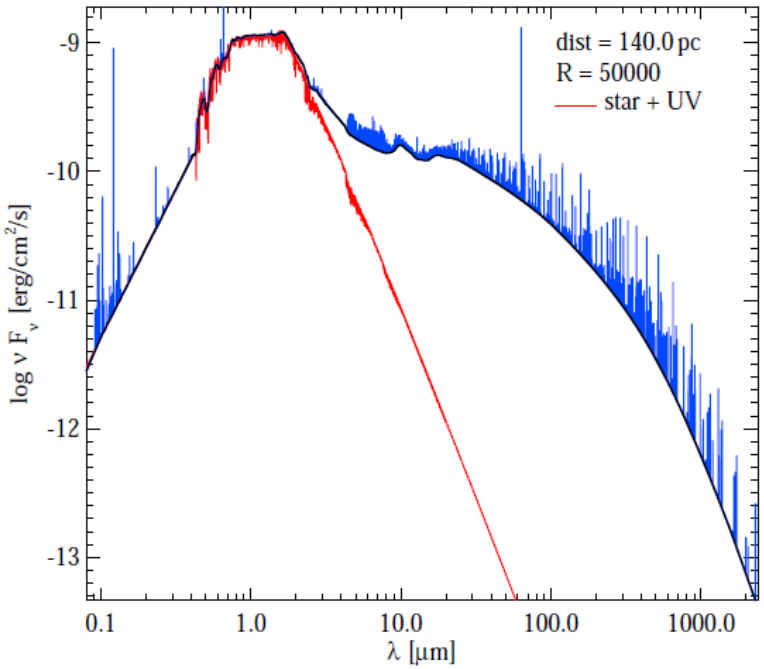


Spectral energy distributions of cool stars computed with the NextGen models.

"New light on dark star", Credit: P. Hauschildt

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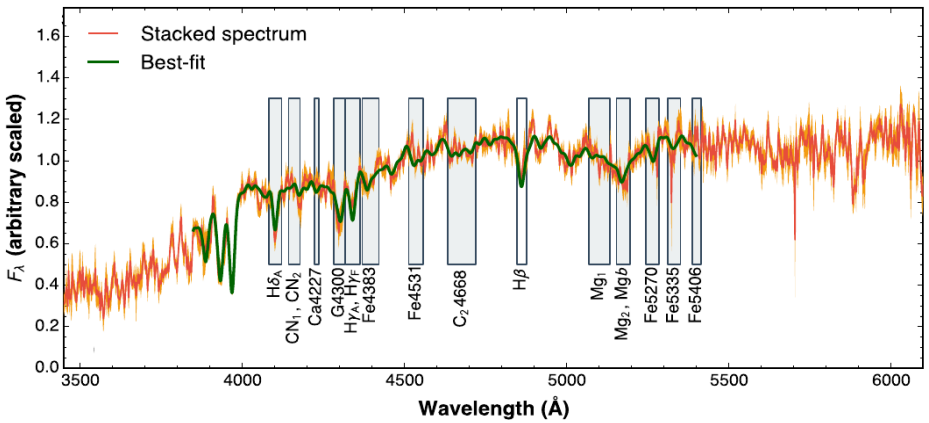


Computed SED of a typical T Tauri star, including spectral lines at high resolution

Credit: DIANA project, University of St Andrews

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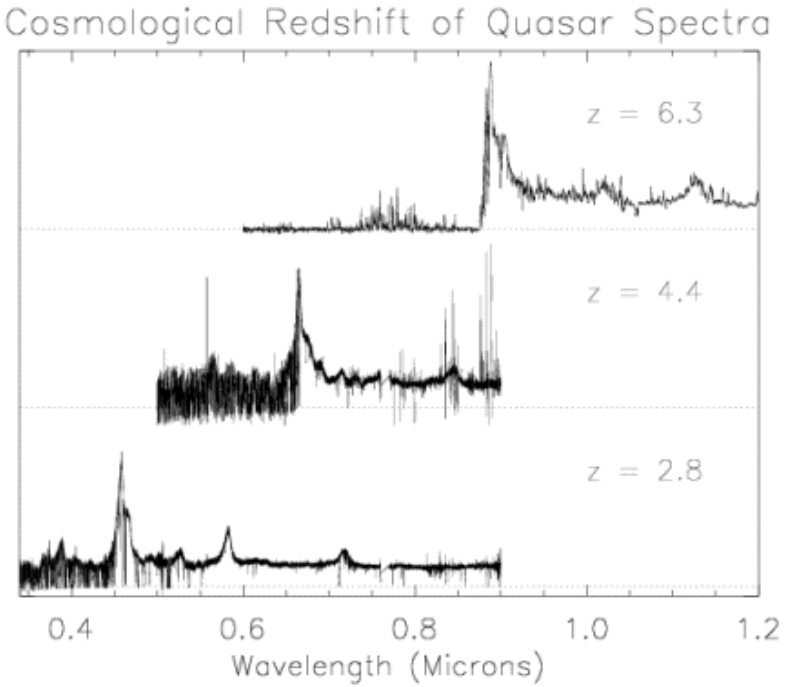
Rest-frame stacked spectra of galaxies at $z \sim 1.6$

Maiolino & Mannucci, 2019,

The Astronomy and Astrophysics Review, Volume 27

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Credit: R. Simcoe

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Astronomical Photometry : origins

- Hipparcos classified stars by brightness
  - 1 (brightest) to 6 (faintest)
- 1856: Pogson
  - m=6: 1/100 of the brightness of an m=1 star

➥ defines logarithmic scale based on $\sqrt[5]{100} = 2.512$

  - visual magnitude based on human eye
  - photographic magnitude based on photographic emulsions

Sun	-26.8
full Moon	-12.7
brightest Venus	-4.8
Sirius	-1.4
Vega	0.0
Polaris	2.0
Visual limit	6
HST deep field limit	30

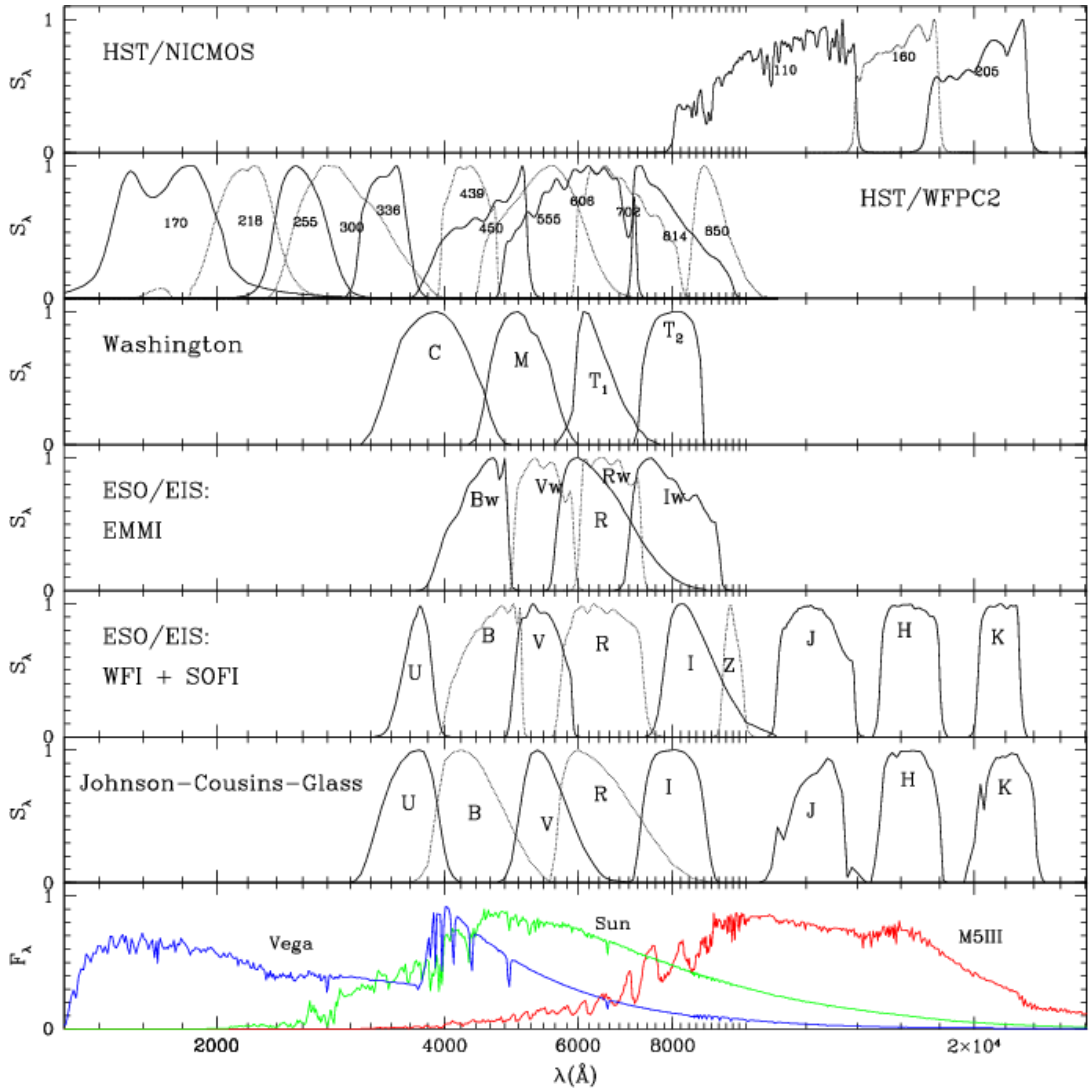
Astronomical Photometry : bolometric quantities

- Total energy radiated by a star :
  - $L_{\rm bol}$ (W) : luminosity
- Flux density collected at distance d from the star :
  - $F_{\rm bol} = \dfrac{L_{\rm bol}}{4\pi d^2}$ (W/m^2) : irradiance (FR: éclairement)
- Apparent bolometric magnitude :
  - $m_{\rm bol} = -2.5 \log_{10}\left(\dfrac{L_{\rm bol}}{4\pi d^2}\right) + C = -2.5 \log_{10}\left(\dfrac{F_{\rm bol}}{F_0}\right)$
  - The "zero" of the magnitude scale is defined by the constant C or equivalently by the constant F_0

Astronomical Photometry : filters

- Actual measurements are done through standardized filters
  - Visible : Johnson-Cousins system : U, B, V, R_c, I_c
  - Visible : Gunn/SDSS system : u, g, r, i, z
  - nIR : 2MASS : JHK_s
- Flux reference :
  - Vega system : m_F(Vega)=0.03 for any filter F (Johnson-Cousins)
  - AB : $m_{\mathrm{AB}} = -2.5 \log_{10} \left(\frac{F_\nu}{F_{\nu,0}} \right)$ with $F_{\nu,0} = 3.631 \times 10^{-23}\ \mathrm{W.m^{-2}.Hz^{-1}}$
- In practice...
  - comparison with "standard" stars
  - magnitudes are extrapolated above the atmosphere
  - CCD image ➔ photon counts ➔ calibrated magnitude

Astronomical Photometry : filters



Filter+detector transmission curves S_λ for various systems re-normalized to their maximum value of S_λ.

The bottom panel presents the spectra of Vega (A0V), the Sun (G2V), and a M5 giant, in arbitrary scales of F_λ.

Girardi et al., 2002, A&A, 391, 195

Astronomical Photometry : color indices

- Color or color index: mag difference in two filters of increasing wavelength
  - $B - V = m_B - m_V$

➔ independent of d but modified by interstellar absorption

- Colors in Vega system: by def. Vega has all colors equal to 0
  - $B - V > 0$: redder than Vega
  - $B - V < 0$: bluer than Vega
- Bolometric correction: $BC = m_{\rm bol} - m_V$

Table 3.6 *Empirical Main Sequence*

Spectral type	$B-V$	log MM$_\odot$	log RR$_\odot$	log T_e K	M_V	$M_{\rm bol}$	BC
O5	−0.34	1.4	0.9	4.60	−5.5	−8.1	−2.6
B0	−0.30	1.15	0.76	4.475	−4.0	−6.2	−2.2
B5	−0.17	0.64	0.505	4.190	−0.9	−2.1	−1.2
A0	0.00	0.324	0.262	3.974	1.5	1.3	−0.2
A5	0.13	0.222	0.185	3.911	2.30	2.33	0.03
F0	0.35	0.122	0.117	3.844	3.33	3.33	0.00
F5	0.45	0.067	0.072	3.810	3.91	3.90	−0.01
G0	0.61	0.006	0.018	3.772	4.63	4.57	−0.06
G5	0.66	−0.015	−0.026	3.762	4.85	4.75	−0.10
K0	0.85	−0.102	−0.083	3.715	5.83	5.63	−0.20
K5	1.20	−0.264	−0.227	3.633	7.60	7.19	−0.41
M0	1.4	−0.366	−0.339	3.604	9.50	8.06	−1.4
M5	1.6	−0.57	−0.54	3.513	13.0	9.98	−3.0
M8	1.8	−0.95	−0.96	3.418	19.5	12.93	−6.6

- Interstellar reddening
  - $A_V = 3.1\, E_{B-V}$ is representative for ISM

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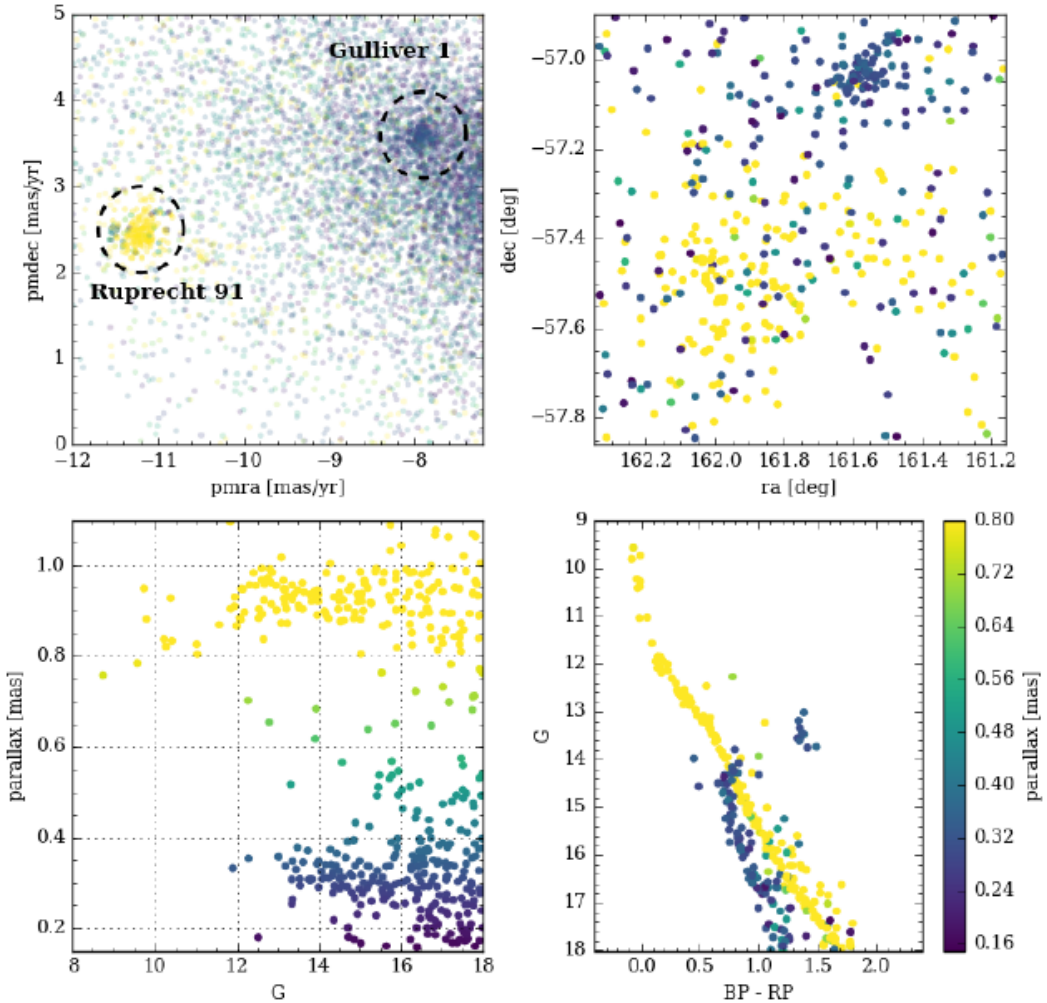
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Astronomical photometry: applications

- Absolute photometry

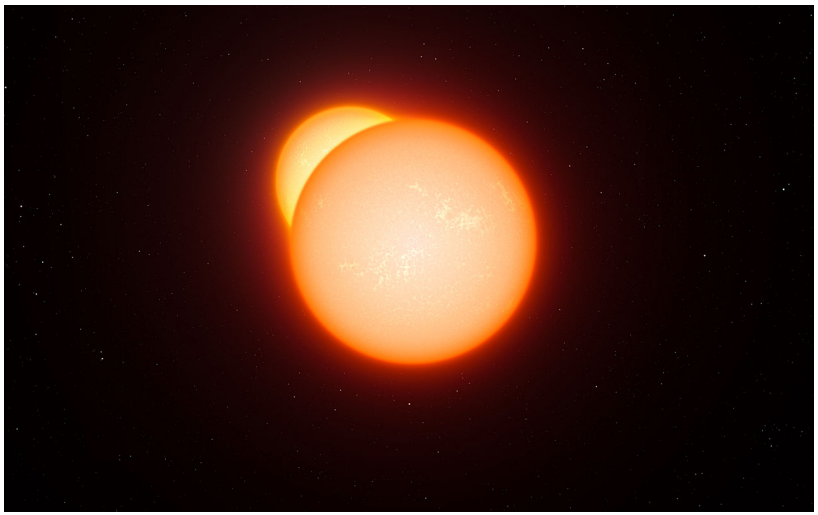


Identification of open clusters for studies of galactic dynamics and stellar evolution

Cantat-Gaudin et al., 2018, Astronomy & Astrophysics

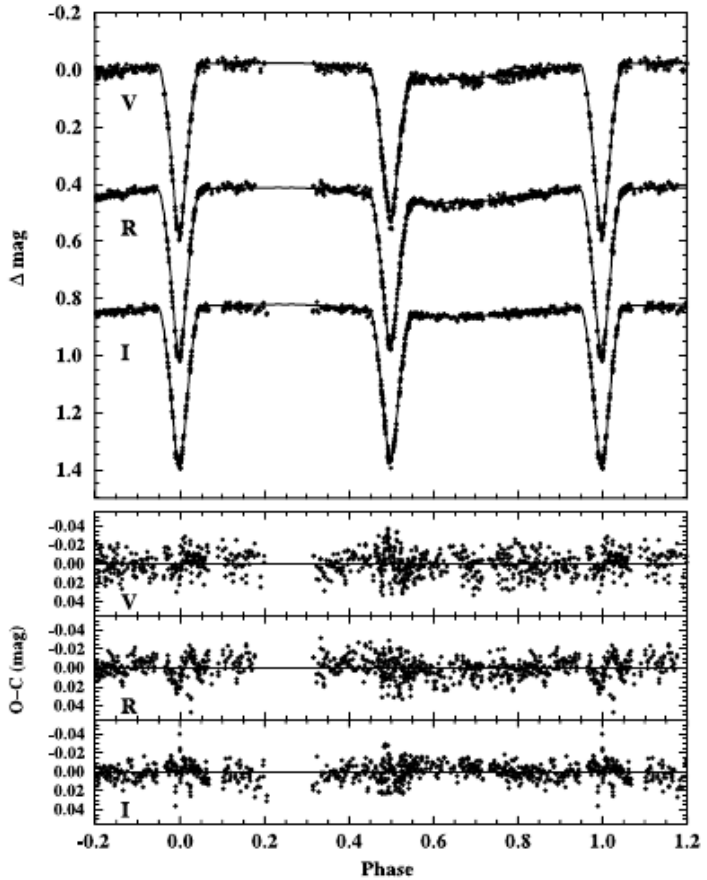
Astronomical photometry: applications

- Differential photometry



Artist view of an eclipsing binary system

Credit: ESO/L. Calçada



Photomety
of the eclipsing binary YY Gem

Torres & Ribas, 2002, Astronomy & Astrophysics

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- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod ISBN-13: 978-2100576401
  - Chapitre 4 : "Connaître les étoiles"
- *"Les étoiles et le Milieu Interstellaire"*, R. Monier, 2006, Ellipses ISBN-13: 978-2729827861
  - Chapitre 1 : "Propriétés observationnelles des étoiles"
- *"Fenêtres sur l'Univers"*, Observatoire de Paris `https://media4.obspm.fr/public/ressources_lu/`
  - Chapitres "Systèmes binaires et multiples" "Température"
- *"Introduction to astronomical photometry, 2nd edition"*, E. Budding & O. Demircan, 2007, Cambridge University Press
- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences
- *"Planets, Stars and Stellar Systems - Volume 2: Astronomical Techniques, Software, and Data"*, Eds T.D. Oswalt & H.E. Bond, 2013
  - Chapitre 1 : "Astronomical Photometry"